

XOR-bitant!

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The *exclusive or* (XOR) function is a binary operation which, when given two true or false values, outputs “true” if and only if the two values are distinct. Replacing “false” and “true” with 0 and 1 respectively, the XOR operation, represented by the operator $\oplus$, can be defined as

$$a \oplus b = a + b \pmod{2}$$

where a and b take on some value from $\{0, 1\}$.

We can also extend this in a way that encompasses inputs beyond just zeros and ones to get the *bitwise XOR* operation ($\oplus$): Given two binary strings, for each pair of corresponding digits, use the XOR operator on the pair of numbers to get the respective digit in the final result, using leading zeros as necessary. For example,

$$\begin{array}{r} 101001_2 \\ \oplus 010111_2 \\ \hline = 111110_2. \end{array}$$

In the spirit of XOR, let’s describe a binary operation $\oplus^m$, which acts on numbers a and b between 0 and $m - 1$:

$$a \oplus^m b = a + b \pmod{m}$$

This operation can be extended to arbitrary inputs base m in a similar way as with XOR; take corresponding digits a and b , and use $a \oplus^m b$ as the digit in the resulting base- m number. In a similar fashion to the bitwise XOR operator, let’s call this operation $\oplus^m$. To demonstrate this operation,

$$\begin{array}{r} 64523 \\ \oplus^9 01422 \\ \hline = 65045. \end{array}$$

1. Show that there is a unique number e base m such that

$$a \oplus^m e = e \oplus^m a = a.$$

2. Is there always a number a' such that $a \oplus^m a' = e$?

3. One day, a mathematician accidentally writes $\oplus^m$ as $\otimes^m$. Then, her friend, reading her work, sees $a \otimes^m b$ and interprets it to mean

$$a \times b \pmod{m}$$

when a and b are between 0 and $m - 1$, with the analogous extension to arbitrary base m numbers denoted by $\otimes^m$.

Does $\otimes^m$ satisfy the same properties as $\oplus^m$? If not, are there any restrictions we can impose on a , b , and/or m for which these properties are satisfied?

4. Do $\oplus^m$ and $\otimes^m$ satisfy the distributive property? That is, does

$$a \otimes^m (b \oplus^m c) = (a \otimes^m b) \oplus^m (a \otimes^m c)?$$